SUMO ROBOT COMPETITION

HANDBOOK

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SECTION 1

INTRODUCTION AND OVERVIEW

The E-JUST Sumo Robot Competition tests engineering ingenuity, control design, sensor integration and mechanical design. Two robots face off inside a circular ring (the Dohyo). The objective is to push, ram or otherwise displace the opposing robot out of the ring. The last robot remaining inside the Dohyo wins the round.

Both autonomous and manual wireless control modes are permitted and treated equally without distinction. Wired (tethered) control is strictly prohibited. The tournament awards 1st, 2nd, and 3rd place from knockouts, alongside a 100-point special prize evaluated by three independent judges (Section 14).

SECTION 2

DEFINITIONS

TERM	MEANING
Dohyo	The circular sumo ring on which matches are fought.
Match	A contest between two teams and their robots, played as best of 3 rounds (first team to win 2 rounds wins the match).
Round	One fight, lasting at most 2 minutes, won by pushing the opponent out or via the bottle tiebreaker challenge (Section 10.2).
Shikiri-Sen	The two start lines on the Dohyo behind which the robots are placed.
Homologation	Technical inspection of a robot before it may compete (Section 7).
Safe area	The area marked by the organizers where operators and assistants stand while a round is running.

SECTION 3

TEAMS, PARTICIPANTS AND CONDUCT BEFORE MATCHES

3.1 Team composition

- A team consists of 4 to 6 members in total: regular team members and one team leader.
- Recording a match is allowed and may be done by any team member. The referee's decision does not depend on recordings.

3.2 Team room, schedule and lateness

- Robots are handed to the organizers at the beginning of the event for inspection (Section 7). Teams attend the rest of the event's sessions while their robots are inspected, and the robots are returned to the teams before the competition starts.
- Once the competition starts, teams stay in the room reserved for them (marked on the area map). Teams leave only when called by an official or volunteer to the waiting area near the game area.
- After a match, the team returns to its reserved room.
- Each team is responsible for following the starting grid (schedule) displayed on the board.
- A team called to play that does not arrive within 5 minutes loses the match.

3.3 Robot number

Each robot displays a number issued by the organizers (found in the Team Folder) on its casing, in a place visible to the referee. The referee uses it to identify the robot.

SECTION 4

ROBOT SPECIFICATIONS

PARAMETER	SPECIFICATION	DETAILS AND RESTRICTIONS
Maximum mass	3000 g (3 kg)	Total mass at the start of a match, measured at inspection.
Start dimensions	Fits inside a 20 cm x 20 cm square	Checked by the organizers at inspection (Section 7), before the competition.
Height	Unlimited	Height is unlimited, but must not exceed the total length (diameter) of the Dohyo (154 cm).
Expansion after start	Allowed	The robot must stay one centralized robot and must not separate into pieces.
Control mode	Autonomous or Manual (wireless remote only)	Wired control prohibited. Both modes treated equally. Mandatory 5-second starting delay applies to all.
Emergency stop	2 buttons required (1 on-robot, 1 wireless)	Two physical kill switches: one on the robot and one wireless remote stop. Independent of code, tested at inspection.
Battery	Secured and inspected	No visibly damaged, swollen or leaking cells; securely mounted.
Chassis durability	Must withstand match contact	Verified by the durability check in Section 7.
Robot number	Displayed	Number issued by the organizers (Section 3.3).

A robot that violates these restrictions loses the match.

SECTION 5

ROBOT RESTRICTIONS

5.1 Permitted

- Impact and ramming mechanisms: rams, wedges, plows and front non-cutting blades, provided their purpose is to displace the opponent. Permitted blades must not be sharp or cutting; their sole purpose is placement at the bottom front to slide under the opponent (movable up and down within a range solely to tilt and push). Normal pushes and bangs are not considered intent to damage.
- **Sensors** of any kind (for example IR distance, line, ultrasonic) used by the robot itself.
- **Flags:** up to two, optional, of any color. Flags are made of fabric, paper or polyester with a plastic pole, stand vertically and stay inside the robot's perimeter. After the robot is placed on the Dohyo it may be touched only to set the flags vertical. Once the match begins the flags may fall.

5.2 Prohibited

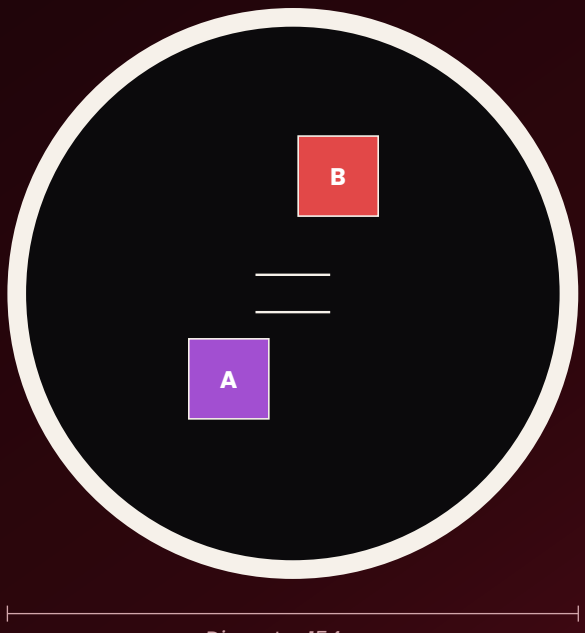
- Parts that could break, scratch or damage the Dohyo.
- Parts intended to damage the opponent's robot or its operator: cutting or sharp blades, spikes, drills, saws, and hammers or piston devices meant to dent or destroy rather than displace.
- Devices that store liquid, powder, gas or other substances for throwing or spraying at the opponent (charged or not).
- Flaming or ignition devices.
- Shooting or throwing devices and projectiles, including any part that separates from the robot and is thrown or launched.
- Sticky substances to improve traction. Wheels and any part touching the Dohyo must not be able to pick up and hold a standard A4 sheet (80 g/m^2) for more than two seconds.
- Suction, vacuum, fans or any part that fixes the robot to the Dohyo or creates artificial downforce (suckers, glue and so on).
- Wave generators or anything that disturbs the opponent's control, such as lasers, flashes or active infrared jamming. Using these sensors normally for the robot's own sensing is allowed; using them to disturb the opponent is not.
- Damaged, swollen or poorly secured batteries.

> **Ram or weapon?**
> All attacking mechanisms must be revised by the organizers, and you are responsible if the robot fails the inspection stage. If you are unsure whether a mechanism counts as a ram or as a weapon, bring a sketch or prototype to the organizers for sign-off before you commit build time to it.

SECTION 6

DOHYO (ARENA) SPECIFICATIONS

FEATURE	SPECIFICATION
Shape and diameter	Circular, 154 cm (1540 mm) in diameter.
Surface color	Black.
Border line	White circular ring, 5 cm (50 mm) wide, around the outer edge.
Start lines (Shikiri-Sen)	Two white lines, each 20 cm long and 3 mm wide, 10 cm apart and centered on the Dohyo. The lines are very thin so they do not confuse edge-detection sensors. Robots may start anywhere behind their own line; starting on the line is not required.
Surface material and thickness	Wood (matte black surface).
Base height	10 cm.



Dohyo layout at the start of a round

- Black surface (Wood, 10 cm base height)
- White border (Tawara): 5 cm wide
- Shikiri-Sen: 2 thin lines, 20 cm long, 10 cm apart, centered on the dohyo
- Robot A: anywhere behind its own line
- Robot B: anywhere behind its own line

SECTION 7

TECHNICAL INSPECTION (HOMOLOGATION)

Every robot must pass inspection before competing. The organizers collect the robots from the teams at the beginning of the event and inspect them themselves while the teams attend the rest of the event's sessions. The robots are returned to the teams before the competition starts. A robot that fails may not enter the Dohyo until it is corrected and re-inspected. Each team gets one re-inspection if the robot fails on weight, dimensions or the durability check.

Checks carried out:

- **Mass** (maximum 3000 g) on the organizers' scale.
- Dimensions: fits 20 x 20 cm start square; height unlimited up to Dohyo diameter (154 cm).
- Emergency stops: 2 required (1 on-robot switch, 1 wireless), independent of code, tested.
- **Battery:** secured, with no visible damage, swelling or leaks.
- **No prohibited devices** (Section 5.2); wheels pass the A4 paper test.
- **Flags,** if used, are correct; the organizer-issued robot number is displayed.
- **Durability check:** any robot with a ramming, wedge, blade or impact mechanism undergoes a brief contact test against a padded test rig. The mechanism and its mounting must withstand match impacts without shedding parts or exposing sharp edges.

Once a robot has passed homologation, it may not be modified until its match ends (Section 11).

SECTION 8

MATCH PROCEDURE

8.1 Officials

Each match is supervised by three referees: a principal referee and two assistant referees. The principal referee's decision on all safety and rule matters is final.

8.2 Format and duration

Each match has 3 rounds (best of three) and is won by the first team to win two rounds. Each round lasts at most 2 minutes.

8.3 Placing the robots

- On the referee's instruction, both teams approach the Dohyo and the operators place their robots at the same time. The referee decides the facing direction of the robots. After placing, robots may not be moved.
- Every part of the robot must be behind its own Shikiri-Sen (start line). The robot may start at any point behind the line; it does not have to be placed on or next to the line. No part may cross the line toward the opponent.
- The referee checks the placement. If it is not correct, the robots are positioned again.

8.4 Start

• On the referee's signal, operators start robots (button or remote). A mandatory 5-second starting delay is required for both autonomous and manual control modes before action begins.
• Movement before the 5-second delay expires gives a restart warning; repeated violations forfeit the round.

8.5 Stop, resume and end

- The match stops and resumes only when the referee announces it. The referee also announces when robots may be placed, when operators retreat to the safe area, and when robots may be taken from the Dohyo.
- The match ends when the referee announces it. Once the robots are retrieved, the decision is final and no objections are accepted.

8.6 Breaks between rounds

There is a break of at most 1 minute between rounds, to clean the robot and its wheels and to select tactics. Repairs, battery changes and reprogramming are not allowed (Section 11).

SECTION 9

SCORING

9.1 When a round is won

- A team legally forces the body of the opposing robot to touch the space outside the Dohyo.
- The opposing robot touches the space outside the Dohyo on its own (drives or falls out unforced).
- One or more wheels of the opposing robot run over the edge of the Dohyo, and the referee counts to three while the robot cannot return onto the Dohyo.
- The opposing robot is damaged by itself (a malfunction not caused by the opponent), cannot continue, and its team representative announces it.

9.2 Winning a match

The first team to win two rounds wins the match (best of three rounds).

SECTION 10

RESTARTS AND STALLS

10.1 Restart conditions

- A robot does not start: one restart. If the same robot fails to start again, the round is awarded to the robot that moves.

- Entangled or orbiting with no perceivable progress for 10 seconds: one restart. If it repeats, the round is decided by the bottle tiebreaker challenge (Section 10.2).
- Both robots stand still without touching for 10 seconds, or move without progress: one restart. If it repeats, the round is decided by the bottle tiebreaker challenge (Section 10.2).
- **Both robots leave the Dohyo at the same time** and it is not clear which touched outside first: the round is replayed. Recordings may be ignored.

10.2 Round time limit and bottle tiebreaker

If the 2 minutes of a round end without a winner, or if a stalled situation repeats after a restart, the round is decided by a tiebreaker challenge. A bottle is placed at the far end of the ring; each robot is placed in the ring individually to hit the bottle in the shortest time. Each robot attempts the challenge individually while the elapsed time is officially calculated. The robot with the shortest time wins the round.

SECTION 11

REPAIRS, MODIFICATIONS AND INTERRUPTIONS

- From the moment a robot passes homologation and is in the waiting area until the match ends (all rounds played), no changes can be made to the robot and no technical timeouts are given.
- No repairs, no battery change or charge, and no reprogramming during a match. Selecting tactics before a round is not considered programming.
- If a front plow, wedge or attachment detaches during a round, it may not be put back. The robot continues without it until the match is over.
- The robot must start and finish the match (all rounds) without modification (except tactics selection) and may not leave the competition area for any reason.
- Battery changes, charging, repairs and replacement of defective parts are allowed only after the match is over and before the next match.
- If a robot is broken during a match and cannot continue, the opponent wins the match. No repair is allowed.

SECTION 12

CONDUCT, DISPUTES AND SAFETY

12.1 Conduct and disputes

- Unsportsmanlike conduct, including tampering with another team's robot, aggressively arguing calls, or deliberately trying to damage an opponent, results in forfeiture of the match.
- Arriving more than 5 minutes late is penalized as stated in Section 3.
- The referee may stop any match immediately if a safety hazard is seen, regardless of match state.

12.2 Safety summary

- Every robot has two accessible emergency stop buttons (one wireless, one on-robot), tested at inspection.
- Batteries are secured and free of visible damage. Unsafe batteries fail inspection.
- Contact may include forceful ramming and displacement, never a mechanism designed to cut, puncture, crush or otherwise damage the opponent.
- Any impact mechanism passes the durability check (Section 7) before competing.
- Operators and assistants stay in the safe area during rounds.

SECTION 13

TOURNAMENT STRUCTURE

The competition is played in **knockout style**. There is no group stage and no round robin. Teams are paired against each other, and the winner of each match advances to the next stage. A team that loses a match is eliminated. The final decides the winner, and a 3rd place match may be played between the two teams that lose their semi-finals.

Every match has 3 rounds (best of three) and is won by the first team to win two rounds (Sections 8.2 and 9.2). The knockout tournament produces three winner places (1st, 2nd, and 3rd place). A 3rd place playoff match will be played between the two teams that lose their semi-finals.

SECTION 14

PRIZES AND JUDGING CRITERIA

Knockout Tournament: Decides 1st, 2nd, and 3rd place through head-to-head match eliminations.
Special Prize (100 Points Total): Awarded across four technical and performance criteria (25 pts each):

1. Best Mechanical Design (25 pts):
Overall mechanical engineering, chassis structure, weight distribution, stability, traction, and efficient use of dimensions/materials. Judged through physical design.

2. Most Competitive Robot (25 pts):
Strongest fighting performance: aggressive engagement, pushing power, recovery from difficult positions, and consistent operation based on match evidence.

3. Best Engineering Innovation (25 pts):
Clever, original technical solution in mechanical design, sensing, control, electronics, or integration. Prioritizes usefulness and execution over complexity.

4. Sportsmanship Award (25 pts):
Respect toward opponents, referees, and organizers, adhering to rules with a positive attitude in victory and defeat.

Evaluation Process: Every team's robot and matches are revised by three independent judges. Each judge fills out a separate scoring sheet out of 100 points (25 points per criterion). A designated official calculates the average of the three judges' points; the team with the highest average score wins the prize.

SECTION 15

CONTACTS

For any question about the competition or this handbook, please contact:

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This handbook is issued by the IEEE RAS E-JUST Student Branch Chapter. The organizing committee may publish clarifications before competition day; during the competition the principal referee's decision is final.